

# Numerov Method for Bound States in the Schrödinger Equation

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## 1 Introduction

The time-independent Schrödinger equation in one dimension is a central equation in quantum mechanics. In dimensionless units ( $\hbar = m = 1$ ), it can be written as

$$\psi''(x) = 2[V(x) - E]\psi(x), \quad (1)$$

where  $V(x)$  is the potential and  $E$  is the energy eigenvalue.

Finding bound states corresponds to solving a boundary value problem where only specific values of  $E$  produce physically acceptable (normalizable) solutions. This project implements a numerical approach based on the Numerov method combined with a shooting technique to compute these eigenvalues and eigenfunctions.

## 2 Numerical Method

### 2.1 Numerov Algorithm

The Numerov method is a finite-difference scheme designed for second-order differential equations of the form

$$\psi''(x) = q(x)\psi(x). \quad (2)$$

It achieves higher accuracy by incorporating information from neighboring points. For a uniform grid with spacing  $h$ , the recurrence relation is

$$\psi_{n+1} = \frac{2(1 + \frac{5h^2}{12}q_n)\psi_n - (1 - \frac{h^2}{12}q_{n-1})\psi_{n-1}}{1 - \frac{h^2}{12}q_{n+1}}. \quad (3)$$

This method is well-suited for the Schrödinger equation due to its stability and accuracy.

### 2.2 Shooting Method

To determine eigenvalues, a shooting method is used. For symmetric potentials:

- Even states satisfy  $\psi'(0) = 0$
- Odd states satisfy  $\psi(0) = 0$

Starting from  $x = 0$ , the wavefunction is propagated outward using the Numerov method. The correct energy eigenvalue is found by requiring the wavefunction to decay at large  $x$ .

A bisection method is used to locate energies where the boundary condition at the edge changes sign.

## 3 Results

### 3.1 Infinite Square Well

The infinite square well provides a benchmark since the exact energies are

$$E_n = \frac{n^2 \pi^2}{8a^2}. \quad (4)$$

The numerical results show excellent agreement with the analytical solution, with relative errors decreasing as the grid is refined.

### 3.2 Harmonic Oscillator

For the harmonic oscillator,

$$V(x) = \frac{1}{2} \omega^2 x^2, \quad (5)$$

the exact energies are

$$E_n = \omega \left( n + \frac{1}{2} \right). \quad (6)$$

The computed eigenvalues match the expected spectrum, and the wavefunctions exhibit the correct number of nodes and symmetry.

### 3.3 Double Well Potential

A quartic double-well potential was also studied:

$$V(x) = ax^4 - bx^2. \quad (7)$$

The results show pairs of nearly degenerate states, corresponding to tunneling between the wells. This behavior has no simple analytical solution and demonstrates the power of the numerical approach.

## 4 Convergence and Accuracy

Two main sources of numerical error were analyzed:

- Grid spacing ( $h$ ): smaller spacing improves accuracy
- Domain size ( $x_{\max}$ ): insufficient domain truncates wavefunctions

The results show systematic convergence as  $h \rightarrow 0$  and  $x_{\max}$  increases, consistent with expectations for finite-difference methods.

## 5 Conclusion

This project demonstrates that the Numerov method, combined with a shooting technique, provides an efficient and accurate way to solve quantum bound-state problems.

The implementation successfully reproduces known analytical results and extends naturally to more complex potentials. The method is robust, easy to implement, and well-suited for computational physics applications.